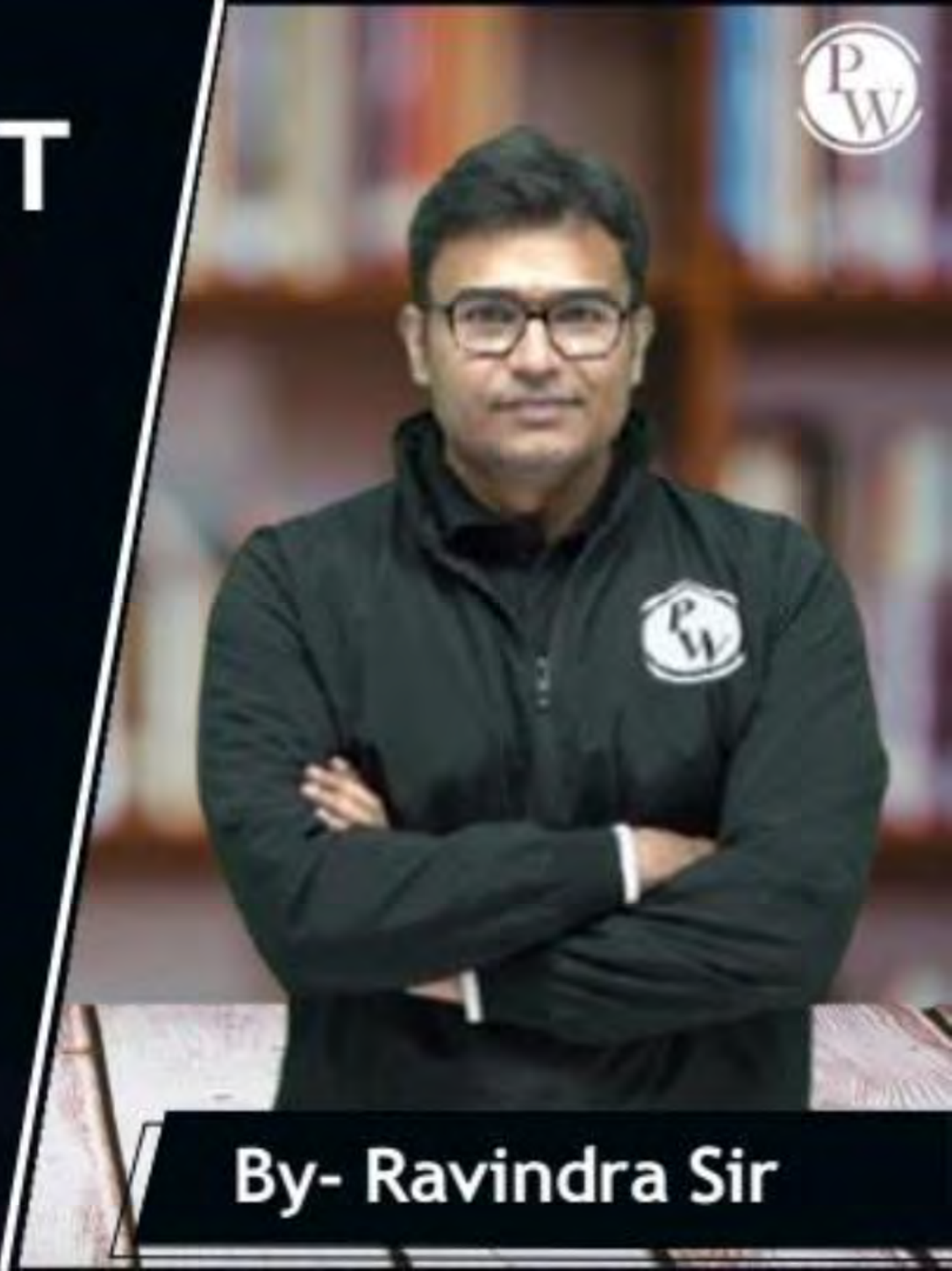
Computer Science & IT

COMPUTER NETWORK
(CN)

Hardware and Various Devices Used in Networking

Lecture No. 02



By- Ravindra Sir

Recap of Previous Lecture



- Topic — DNS
- Topic
- Topic

Topics to be Covered



- Topic
- Topic
- Topic

DNS, SMTP

DNS records :-

DNS servers that together implement DNS distributed database store resource records (RR). Each DNS Reply carries one or more RR.

A resource record is a four tuple that contains following fields

(Name, Value, Type, TTL)

TTL is time to live of a RR. It determines when a resource should be removed from a cache.

RR: (Name, Value, Type, TTL)

A

If type = A, the name is hostname and Value is IP address

(relay1.westcost.yahoo.com, 200.1.2.3, A)

RR : (Name, value, type, TTL)

NS

If type=NS, the name is domain name and the value is host name of the authoritative DNS Server that knows how to obtain the IP addresses for host in the domain.

(yahoo.com, dns.yahoo.com, NS)

RR. (Name, Value, type, TTL)

CNANE

the value is Canonical name of the alias host name NAME.

(yahoo.com, relay1.westcost.yahoo.com, CNAME)

RR (Name, Value, Type, TTL)

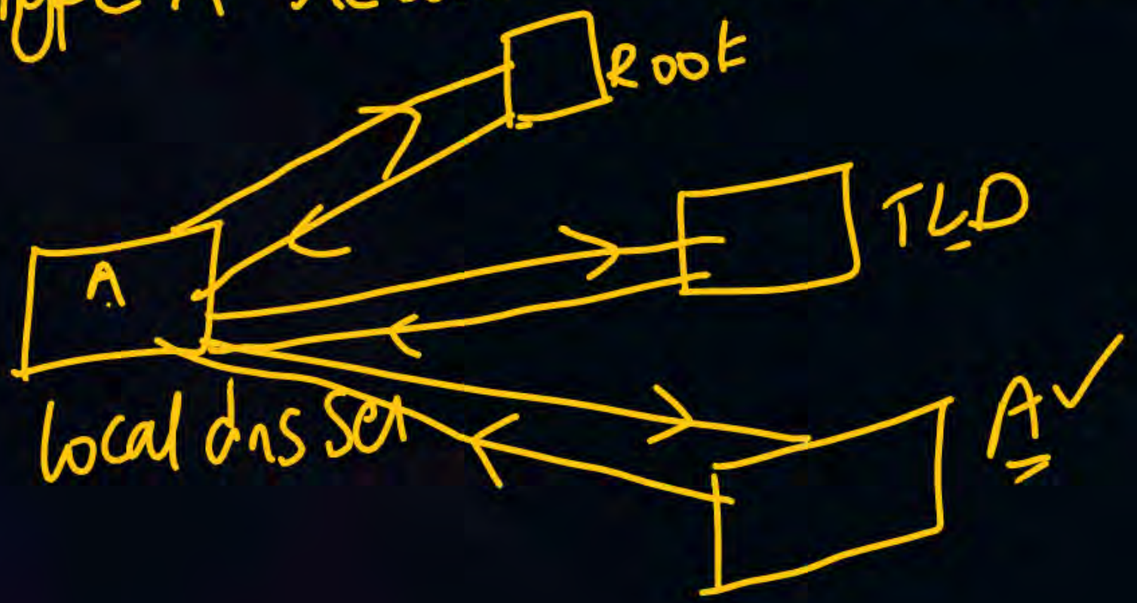
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MX

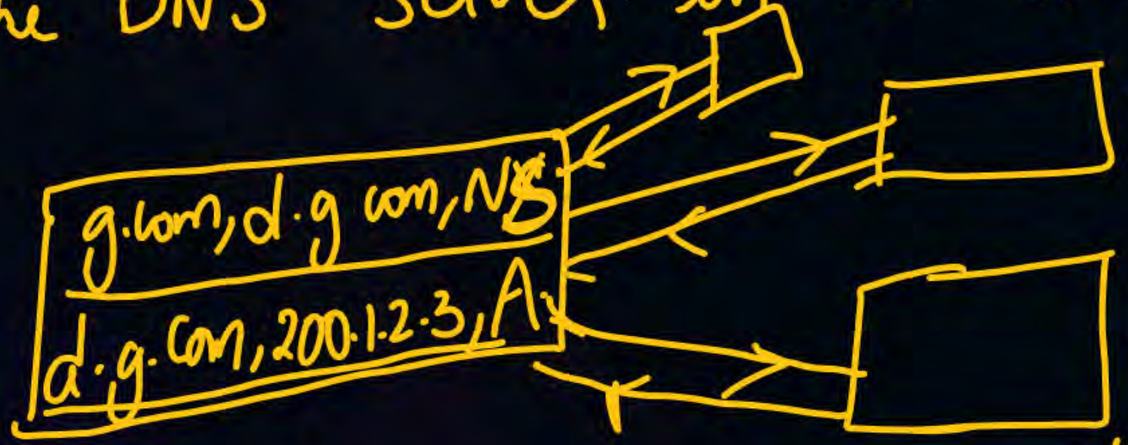
If type = MX, the value is canonical name of the mail Serv

(yahoo.com, mail.yahoo.com, MX)

→ If a DNS Server is authoritative for a particular host name the DNS Server contains Type A record for the host name If the DNS server is not authoritative, then Server will contain Type A record in the cache



If a server is not authoritative server for a host name, then it can contain Type NS for the domain that includes the host name as well as type A record for the IP address of the of the DNS server in the values of the NS record

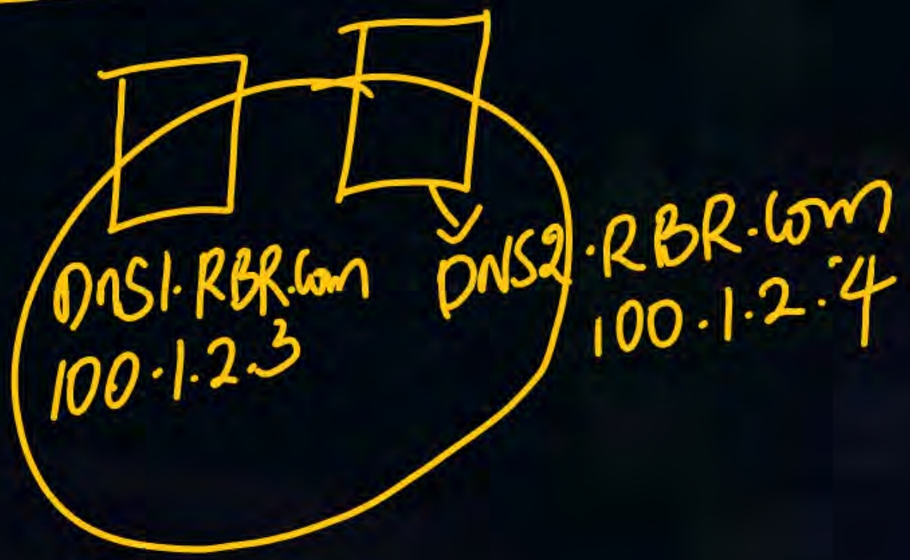


Inserting Records into DNS Server :-

TLD .com

RBR.com, dns1.RBR.com, NS
dns1.RBR.com, 100.1.2.3, A

Registrar ← RBR.com

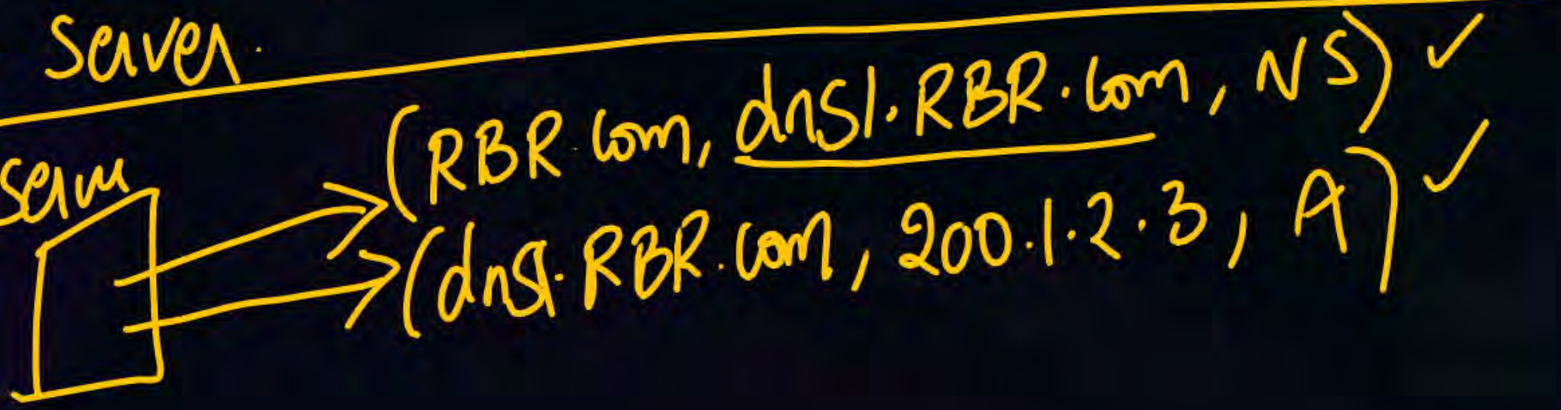


Suppose you have Stated a Start up RBR.
First you will have to register the domain name RBR.com
at a registrar. A registrar is a Commercial entity
that verifies the uniqueness of the domain name,
enters domain name into DNS data
Prior to 1999, a Single registrar, network Solutions used to
have a monopoly for com, net and org domains.
But now there a many registrars Competing for Customers
and Internet Corporation for assigned names and numbers (ICANN)

accredits the various registrars

When you register the domain name RBR.com with a registrar you need to provide the registrar with names and IP addresses of primary and secondary authoritative DNS servers

For each of these authoritative DNS servers, the registrar makes sure type NS and type A records are entered into TLD server.



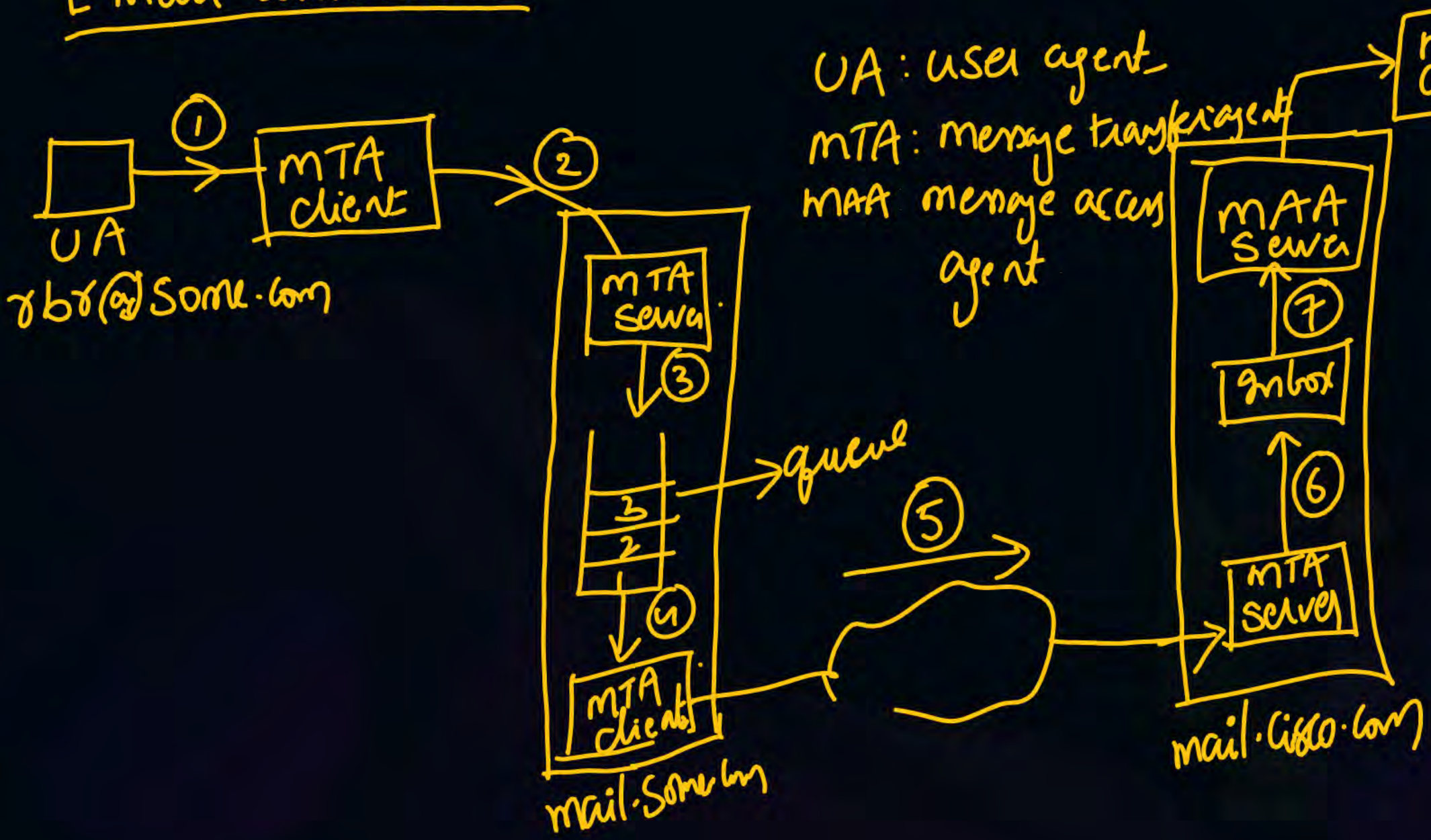
You have to make sure type A resource record for your web-server www.RBR.com and type MX record for your mail server mail.RBR.com are entered in the authoritative DNS server

DNS1.RBR.com

www.RBR.com, 200.12.4, A
RBR.com, mail RBR.com, MX
mail.RBR.com, 200.12.5, A

Authoritative DNS Server

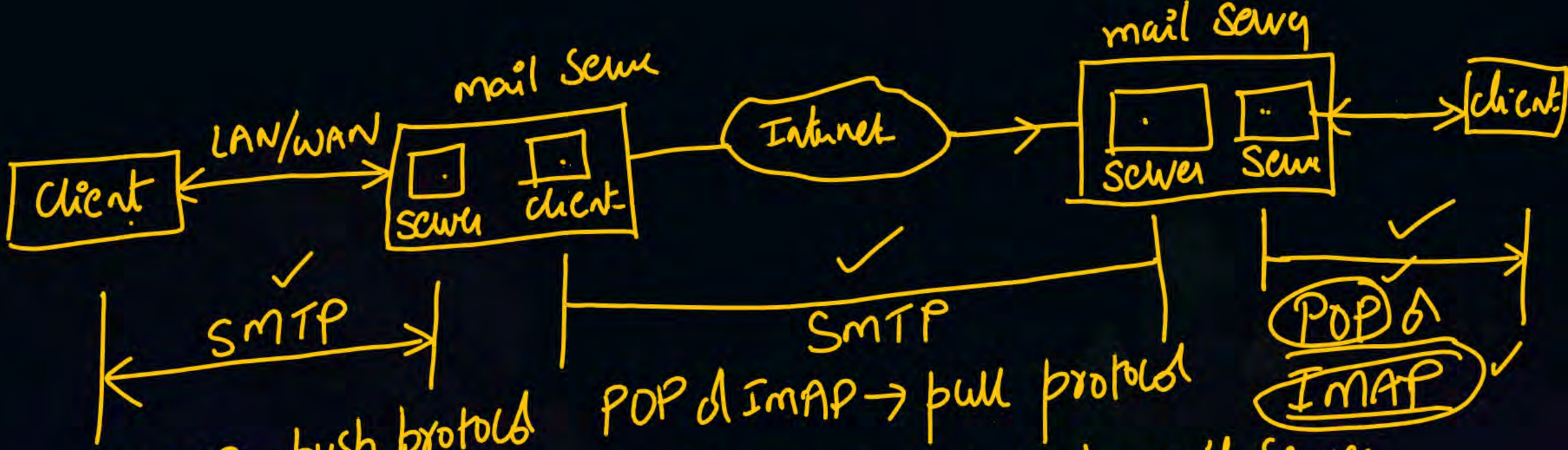
Email architecture:-



User agent :- The first component of an electronic mail system is the user agent. Ex :- MS outlook. It provides services to the user to make the process of sending and receiving a message easily. A user agent is a S/w package that that composes, reads, replies to, and fdwand messages It also handles local inbox on the user computer

message transfer agent (SMTP):-

Port no: 25, it uses TCP



SMTP - push protocol POP & IMAP → pull protocol

SMTP is used two times b/w sender and sender mail server and b/w two mail servers

Currently two message access protocols are available

- POST office protocol, version 3 (POP3)
- Internet mail Access protocol, version 4 (IMAP 4)

SMTP Commands and Responses:-

Keyword	arguments	Description
HELO	Senders host-name	Identifies the sender of the message
MAIL FROM	Sender of the message	Identifies the email of the sender
RCPT TO	Intended Recipient	Identifies the recipient of the message
DATA	Body of the email	Sends the actual message
QUIT		Terminates the message

Keywold	argument	description
RST		aborts the current mail transaction
VRFY	Name of the recipient	Verifies the address of the recipient
NOOP		Checks the status of the recipient
HELP	Command name	Asks the recipient to send information about the command sent as argument
SEND FROM	Intended Recipeent	Specifies that the mail is delivered only to the terminal of the recepient And not to mail box.

SMOL FROM	Intended Recipient	Specifies mail to be delived to the terminal of the mailbox of the Recipient
SMAL FROM	Intended Recipient	Specifies that mail to be delivered to the terminal and mail box of the recipient

mail transfer phases :-

Connection establishment, mail transfer, connection termination

Connection establishment :-

After a client has made TCP connection to the well known SMTP port 25, the SMTP server starts a connection phase

This phase involves 3 steps

1) The server sends code 220 (server is ready) to tell the client that it is ready to receive mail. If the server is not ready, it sends code 421

2) Client sends HELO message to identify itself
This step is necessary to inform the server the domain of the client
3) Server responds with code 250 saying Request Command Completed

Message transfer :-

After connection is established b/w SMTP client and server, a single message b/w a sender and one or more recipients can be exchanged. This phases involves 8 steps

1) The client sends the "MAIL FROM" message to introduce sender of the message It includes mail address of the sender. This is needed to give the server the return mail address for returning errors and reporting messages

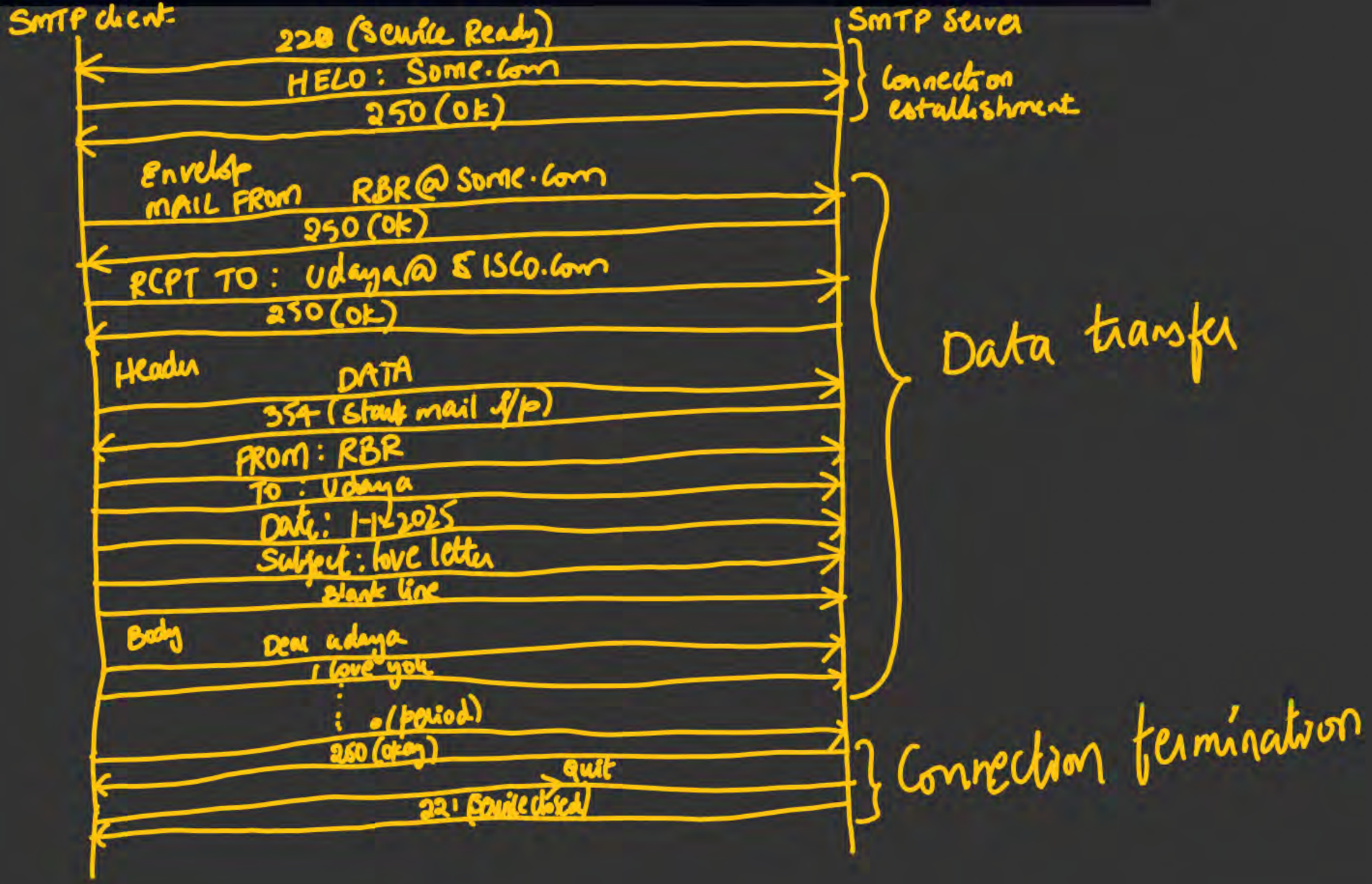
2) Server responds with 250 (OK)
3) The client sends "RCPT TO" message which includes the mail addrss of the recipient
4) The server responds with 250 (OK)
5) The server sends "DATA" menage to initialize the message transfe
6) Server responds with code 354 (Start mail i/p)
7) The client sends the contents of the message in consecutive lines. Each line is terminate with two characters, end of the line token (Carriage return and line feed)
The message is terminated by a line containg just one period (.)

8) Server responds with 250 (OK)

Connection termination :-

After the message is transferred successfully The client closes the connection. This phase has two steps

1) The client sends the QUIT command
2) The server responds with 221


SMTP client
SMTP server
220 (Service Ready)
HELO : Some.com
250 (Ok)
Connection establishment
Envelop
MAIL FROM RBR@Some.com
250 (Ok)
RCPT TO : udaya@ ISCO.com
250 (Ok)
Header
DATA
354 (Start mail i/p)
FROM : RBR
TO : Udaya
Date : 1-1-2025
Subject : love letter
Blank line
Body
Dear udaya
I love you
.
.
.
. (period)
250 (Okay)
Data transfer
quit
221 Service closed
Connection termination

THANK - YOU